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Simulation and experimental validation of workspace computation for mobile cable-driven parallel robots in terrestrial and lunar environments

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Abstract This study aims to quantitatively analyze the workspace characteristics of a mobile cable-driven parallel robot (MCDPR) under varying gravitational conditions and to evaluate its applicability. To this end, we propose a workspace computation algorithm that comprehensively considers tension constraints, tipping prevention, and rotation prevention based on the static equilibrium conditions of the MCDPR. The validity of the proposed algorithm was verified through simulations and experiments, demonstrating strong agreement with theoretical predictions. Subsequently, simulations incorporating lunar gravity and friction conditions were conducted to analyze workspace variations in extraterrestrial environments. The results revealed how reduced gravitational force and friction affected the configuration of the workspace. This study suggests that autonomous construction technology based on MCDPR is feasible for lunar applications and provides foundational data for the design of future space construction platforms.

1. Introduction

With the growing interest in space exploration and the potential for planetary habitation, the demand for automated construction technologies in extreme environments such as the Moon is rapidly increasing. In such settings, the difficulty of human involvement and the limitations in resource utilization necessitate autonomous and efficient construction methods. In response to these challenges, 3D additive construction has emerged as a promising approach for building infrastructure using in-situ resources. Its capability for autonomous operation makes it well-suited for both manned and unmanned construction applications [1–4].

Currently, articulated robotic arms and gantry systems are the primary platforms used for 3D additive construction. Articulated robotic arms are advantageous for fabricating complex-shaped structures due to their high mobility and versatile task execution capabilities. In contrast, gantry systems are well-suited for constructing large-scale structures owing to their wide operational range and high precision. However, robotic arms have limited workspaces, and gantry systems require large installation areas and have low mobility. Moreover, both systems weigh over 3 tons, resulting in transportation costs exceeding \$4000 per kilogram when deployed in space, which poses significant challenges in terms of mass and transport efficiency for space applications [5–11].

As an alternative to overcome these limitations, the mobile cable-driven parallel robot (MCDPR) has been proposed. MCDPR is a parallel-structured system that utilizes multiple cables to fix and control the end-effector in three-dimensional space. Its simple structure and lightweight design make it advantageous in terms of reducing transportation costs. Furthermore, its wide workspace, high payload capacity, and mobility based on mobile platforms make it well-suited for additive construction of structures at various scales [12–14]. MCDPR offers a promising solution to the limited workspace and excessive weight of conventional robotic arms and gantry systems, thereby opening new possibilities for 3D additive construction technologies in space environments.

To apply the MCDPR to space environments, one essential factor must be considered: the difference in gravitational acceleration. For example, the Moon's gravity is only about one-sixth that of Earth. Such a difference can significantly affect the cable tension distribution, frictional interaction with the ground, and the influence of self-weight, even when the robot configuration remains identical. Consequently, the feasible workspace may change substantially. Therefore, quantitatively analyzing how the difference in gravity between the Moon and Earth affects the workspace of the MCDPR is a crucial step in evaluating its applicability to space-based additive construction.

In this study, a workspace computation algorithm based on the static equilibrium of the MCDPR was proposed and implemented in CoppeliaSim. The algorithm was experimentally validated under terrestrial conditions using a prototype MCDPR with tension control. The verified simulation environment was then adapted to lunar gravity and friction conditions to quantitatively analyze how reduced gravity and friction alter the MCDPR's workspace. Through this analysis, the study aims to assess the feasibility of MCDPR-based 3D construction systems for lunar applications and to provide foundational data for designing robot-based construction platforms suitable for extra-terrestrial environments with varying gravitational conditions.

2. Structure of the MCDPR

Since cables can only apply pulling forces to the end-effector, controlling n degrees of freedom requires at least $n+1$ cables [15, 16]. In this study, eight cables were adopted to control all six degrees of freedom of the end-effector. Furthermore, to expand the workspace in the XY plane, four mobile platforms were employed. Each mobile platform is a differential drive robot equipped with two drive motors and one caster wheel. The structure of the prototype MCDPR is illustrated in Fig. 1.

The control length of each cable is defined by Eq. (1) [14].

$$l_i = \|\vec{a}_i - \vec{p}_{EE} - R_{EE} \cdot \vec{b}_i\| \quad (1)$$

Here, l_i denotes the length of the i -th cable, $\vec{a}_i$ is the position vector of the i -th pulley, $\vec{b}_i$ is the position vector of the anchor point where the i -th cable is connected to the end-effector, $\vec{p}_{EE}$ is the target position vector of the end-effector, and R_{EE} is

the rotation matrix representing the target orientation of the end-effector.

3. Workspace computation algorithm

To fix and control the end-effector at a specific position, the cable tensions must counterbalance the external forces acting on the end-effector to achieve static equilibrium. The set of tensions that satisfies the static equilibrium of the end-effector is given by Eq. (2) [17].

$$T = T_p + T_h = -W^+ \cdot F + (I - W^+ \cdot W) \cdot T_{ref} \quad (2)$$

Here, T is the matrix of cable tensions that satisfies the static equilibrium of the end-effector; T_p is the particular tension matrix, and T_h is the homogeneous tension matrix. F denotes the external force matrix acting on the end-effector. W is the wrench matrix, which consists of vectors representing the forces and moments generated by the cables, and W^+ is the Moore–Penrose pseudoinverse of W . T_{ref} is the reference tension matrix selected by considering physical constraints [17].

The workspace of the MCDPR refers to the spatial region where the end-effector can be positioned and controlled. Specifically, it is defined as the set of locations where a tension set satisfying the static equilibrium condition in Eq. (2) exists, while simultaneously meeting the cable tension condition, the tipping condition of the mobile platforms, and the rotating condition of the mobile platforms. When determining the workspace, the key criterion is whether a set of cable tensions exists that satisfies the static equilibrium of both the EE and the mobile platforms. Although dynamic factors such as cable elongation or vibration can influence transient responses during control, they do not affect the existence of feasible tension sets. Therefore, these factors were not considered in this study. This is because the workspace is evaluated under a quasi-static assumption, where static equilibrium is presumed to hold at each posture of the EE during its motion.

3.1 Tension condition

The tension constraints in the MCDPR specify lower and upper bounds on cable forces that keep the cables taut while preventing cable failure and damage to MCDPR components, as defined in Eq. (3) [13].

$$T_{min} < T < T_{max} \quad (\text{element-wise}) \quad (3)$$

Here, T_{min} denotes the minimum tension required to prevent the cable from sagging, and T_{max} represents the maximum allowable tension set to avoid damage to the MCDPR.

3.2 Tipping condition

In the case of a conventional CDPR, the system frame is firmly anchored to the ground or a structure, so it is generally

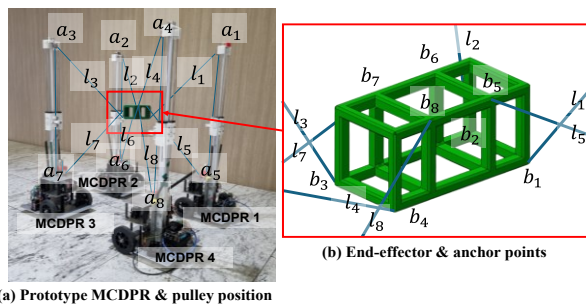


Fig. 1. The structure of the MCDPR.

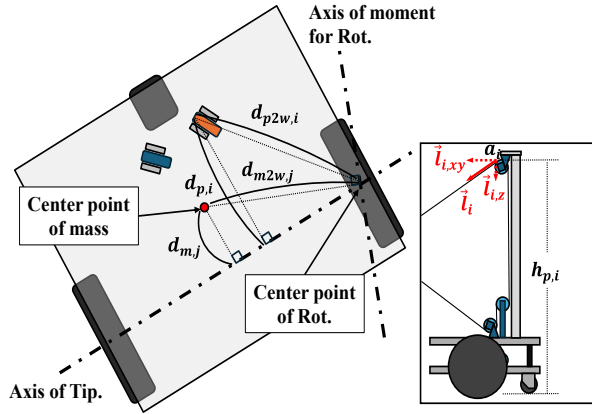


Fig. 2. Geometric parameters for evaluating tipping and rotating conditions of a mobile platform in the MCDPR system.

sufficient to consider only the tension condition. However, since the MCDPR gains mobility through the use of mobile platforms and is no longer fixed to the ground, ensuring the static equilibrium of each mobile platform becomes essential. One such condition necessary for maintaining static equilibrium is the tipping condition, which prevents the MCDPR from tilting. Tipping occurs when the moment generated by the cable tensions about an axis on the contact plane of the wheels exceeds the moment generated by the MCDPR's own weight. Tipping typically arises around the line segment connecting the contact points of the wheels. In this study, considering the range of cable actuation, the tipping condition is defined with respect to the line segment connecting the two drive wheels, as illustrated in Fig. 2. The condition is expressed by Eq. (4).

$$M_{tm,j} > M_{tc,j} + M_{tc,j+4} \quad (4)$$

Here, $M_{tm,j}$ represents the moment generated by the weight of the j -th robot, which can be calculated using Eq. (5).

$$M_{tm,j} = -m_j \cdot g \cdot d_{m,j} \quad (5)$$

Here, m_j denotes the mass of the j -th robot, g is the gravitational acceleration (with $1G = -9.81 \text{ N/kg}$ on Earth), and $d_{m,j}$ is the distance between the center of mass of the j -th robot on the contact plane and the line segment connecting the contact points of the two drive wheels.

$M_{tc,i}$ is the moment generated by the tension of the i -th cable, and it can be calculated using Eq. (6).

$$M_{tc,i} = (T_i \cdot \vec{l}_{i,z}) \cdot d_{p,i} + (T_i \cdot |\vec{l}_{i,xy}|) \cdot \cos(\gamma_{c,i} - \gamma_{m,j}) \cdot h_{p,i} \quad (6)$$

This equation decomposes the cable tension into vertical (z -component) and horizontal (xy -component) directions relative to the contact plane and calculates the moment generated by each component. Here, T_i denotes the tension in the i -th cable; $\vec{l}_{i,z}$ is the z -component of the unit direction vector of the i -th cable,

and $\vec{l}_{i,xy}$ is the xy -component. $\gamma_{c,i}$ represents the yaw angle of the i -th cable, and $\gamma_{m,j}$ is the yaw angle of the j -th robot. $d_{p,i}$ denotes the distance between the i -th pulley and the line segment connecting the two drive wheels, while $h_{p,i}$ is the height of the i -th pulley.

3.3 Rotating condition

Similar to the tipping condition, the rotating condition must also be considered to ensure the static equilibrium of the mobile platforms.

Rotating refers to the phenomenon where the robot rotates around an axis perpendicular to the contact plane due to wheel slippage. It typically occurs around one of the contact points, where the supporting force becomes concentrated near a wheel aligned with the cable direction, as illustrated in Fig. 2. As a result, the supporting forces on the other wheels decrease, leading to reduced friction, which can no longer counteract the horizontal moment generated by the cable tensions. To define the rotating condition that prevents such rotation, it is first necessary to calculate the supporting force acting on the wheel that is not the main point of support. When the support is concentrated near the cable direction, this point is referred to as the dominant support point. The rotating moment is then calculated with respect to the axis on the contact plane perpendicular to the line connecting this support point and the robot's center of mass. The total moment, which includes both the moment generated by the cable tensions and the moment due to the robot's weight, is given by Eq. (7).

$$M_{r,j} = M_{rm,j} + M_{rc,j} + M_{rc,j+4} \quad (7)$$

Here, $M_{r,j}$ denotes the total moment used to evaluate support concentration for the j -th robot, and $M_{rm,j}$ is the moment generated by the weight of the j -th robot, which is calculated using Eq. (8).

$$M_{rm,j} = -m_j \cdot g \cdot d_{m2w,j} \quad (8)$$

Here, $d_{m2w,j}$ represents the distance between the center of mass of the j -th robot on the contact plane and the contact point of the wheel where the supporting force is concentrated.

$M_{rot,cable,i}$ represents the moment generated by the tension of the i -th cable, and it is calculated using Eq. (9).

$$M_{rc,i} = ((T_i \cdot \vec{l}_{i,z}) \cdot d_{p2w,i} - (T_i \cdot |\vec{l}_{i,xy}|) \cdot h_{p,i}) \cdot \cos(\gamma_{c,i} - \gamma_{m2w,j}) \quad (9)$$

Here, $d_{p2w,i}$ denotes the distance on the contact plane between the position of the i -th pulley and the contact point of the wheel where the supporting force is concentrated. This equation computes the moment by decomposing the cable tension into vertical (z) and horizontal (xy) components relative to the contact plane, and then projecting these components onto the rotation axis.

The moment calculated using Eq. (7) represents the remaining moment after the moment due to the robot's weight is counteracted by the moment generated by the cable tensions. If this remaining moment is assumed to result from a virtual force acting at the center of mass, the magnitude of this virtual force can be calculated using Eq. (10).

$$F_{r,j} = \frac{M_{r,j}}{d_{m2w,j}} \quad (10)$$

The virtual force obtained from Eq. (10) must be supported by the wheels that are not the primary support points. Based on the relationship between this virtual force and the support forces of the non-dominant wheels, the support force of the non-dominant drive wheel can be calculated using Eq. (11).

$$F_{n,j} = F_{r,j} - F_{c,j} \quad (11)$$

Here, $F_{n,j}$ denotes the support force of the non-dominant drive wheel of the j -th robot, and $F_{c,j}$ represents the support force acting on the caster wheel of the j -th robot. In this study, it is assumed that the caster wheel is always outside the dominant support region within the cable actuation range.

Additionally, $F_{c,j}$ must counteract the moment generated by $F_{r,j}$ about the central axis defined by the line segment connecting the two drive wheels. Therefore, $F_{c,j}$ can be calculated using Eq. (12).

$$F_{c,j} = \frac{F_{r,j} \cdot d_{m,j}}{d_{c,j}} \quad (12)$$

Here, $d_{c,j}$ denotes the distance on the contact plane between the contact point of the caster wheel and the line segment connecting the contact points of the two drive wheels of the j -th robot.

Since the caster wheel is not constrained with respect to the contact plane, it cannot generate frictional forces to resist rotational motion. Therefore, the frictional force that prevents rotating must rely solely on the support force of the non-dominant drive wheel, and the corresponding moment can be expressed as in Eq. (13).

$$M_{f,j} = \mu F_{n,j} \cdot d_{w2w} \quad (13)$$

Here, μ denotes the coefficient of friction between the wheels and the ground, and d_{w2w} is the distance between the two drive wheels.

Finally, the rotating condition is defined as shown in Eq. (14).

$$M_{f,j} > M_{yc,j} + M_{yc,j+4} \quad (14)$$

Here, $M_{yc,i}$ is the moment generated by the i -th cable about the axis perpendicular to the contact plane, and it is calculated using Eq. (15).

$$M_{yc,i} = T_i \cdot |\vec{l}_{i,xy} \times \vec{d}_{p2w,i}| \quad (15)$$

Here, $\vec{d}_{p2w,i}$ represents the distance vector on the contact plane between the i -th pulley and the support point of the wheel where the supporting force is concentrated.

3.4 Workspace computation algorithm for the MCDPR

The workspace of the MCDPR is determined by calculating the set of cable tensions for each point in space using Eq. (2) and then verifying whether the resulting tension set satisfies the tension condition, tipping condition, and rotating condition. Unlike conventional approaches that use a fixed T_{ref} , this study incrementally increases T_{ref} starting from zero and compares the resulting tension sets against the constraints. If a computed tension set satisfies all conditions, the corresponding point is included in the workspace. If any condition is violated and the tension exceeds the maximum allowable limit, that point is excluded. In summary, only those positions that satisfy static equilibrium and all three conditions (tension, tipping, and rotating) are defined as part of the workspace. This process is described in Algorithm 1.

4. Workspace of the MCDPR in earth environment

4.1 Theoretical workspace computation

To analyze the influence of each condition on the workspace, separate workspace calculations were performed under four scenarios: considering only the tension condition; considering both the tension and tipping conditions; considering both the tension and rotating conditions; and considering all conditions

Algorithm 1. Workspace computation algorithm.

```

1: Initialize homogeneous tension:
    $T_{ref} \leftarrow \text{zeros}(8,1)$ 
2: Initialize tension set:
    $T \leftarrow \text{zeros}(8,1)$ ,  $workspace \leftarrow \text{False}$ 
3: while  $T < T_{max}$  do
4:    $T = W^+ F + (I - W^+ W) T_{ref}$ 
5:   if  $T > T_{min}$  then
6:     if  $M_{m,j} > M_{ic,j} + M_{ic,j+4}$  then
7:       if  $M_{f,j} > M_{yc,j} + M_{yc,j+4}$  then
8:          $workspace \leftarrow \text{True}$ 
9:         Break()
10:      end if
11:    end if
12:  end if
13:   $T_{ref} \leftarrow T_{ref} + c \cdot \text{ones}(8,1)$ 
14: end while
15: if  $workspace$  is True then
16:   return  $workspace, T$ 
17: else
18:   return  $workspace$ 
19: end if

```

Table 1. 3D positions of the pulleys used in the MCDPR prototype.

	a_1	a_2	a_3	a_4	a_5	a_6	a_7	a_8
x [mm]	348.3	-350.4	-354.0	349.3	376.6	-376.9	-376.9	376.6
y [mm]	285.8	286.7	-287.1	-285.8	313.6	313.7	313.4	313.6
z [mm]	274.0	274.0	273.7	274.0	1034.3	1035.7	1035.0	1035.6

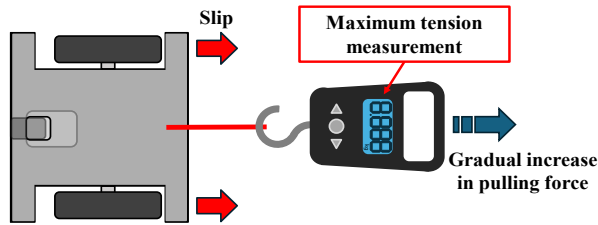


Fig. 3. Schematic diagram of friction coefficient measurement.

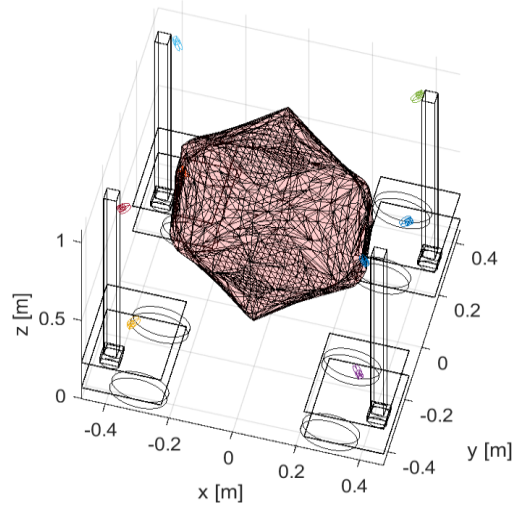


Fig. 4. Workspace based on tension condition.

simultaneously. The kinematic configuration was based on the actual prototype structure of the MCDPR. The spacing between mobile platforms was set to 60 cm, and the vertical distance between the upper and lower pulleys was 81 cm. The height from the ground to the top surface of the mobile platform is 25.1 cm. As a result, the positions of the pulleys are configured as shown in Table 1.

Each robot weighed approximately 8.2 kg, and the end-effector weighed 190 g. The end-effector was modeled as a cube with an edge length of 8 cm. The coefficient of friction between the wheels and the ground was measured using a digital scale with a 5 g resolution, as shown in Fig. 3, and was determined to be 0.237.

The resulting workspaces under each condition are illustrated in Figs. 4-7.

Fig. 4 illustrates the workspace computed considering only the tension condition, resulting in the widest coverage. However, this may overestimate the actual workspace of the MCDPR, as

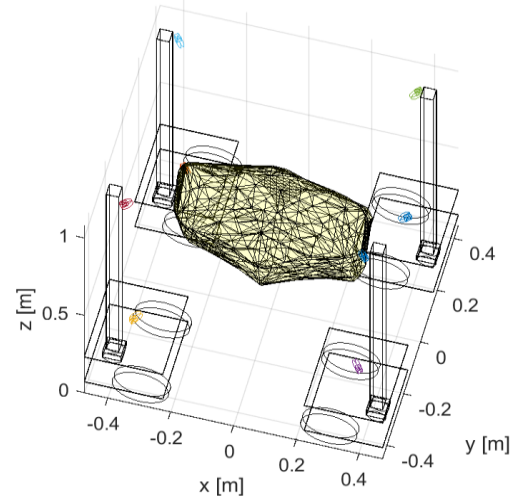


Fig. 5. Workspace based on tipping condition.

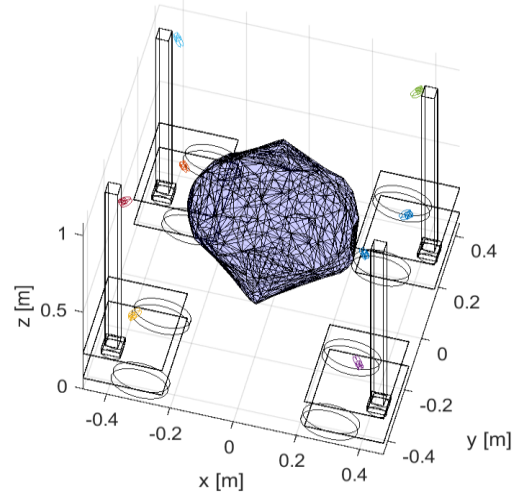


Fig. 6. Workspace based on rotating condition.

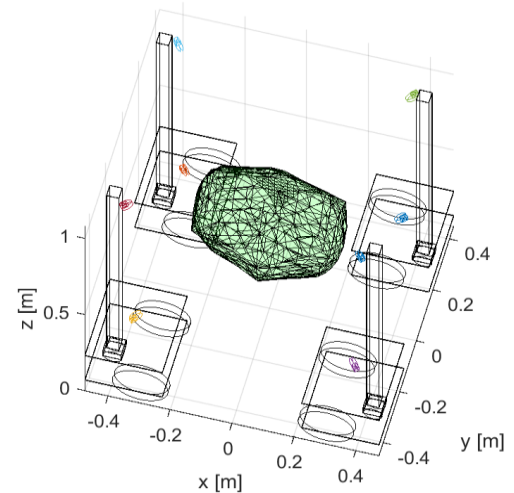


Fig. 7. Workspace based on all condition.

the static equilibrium of the mobile platforms is not accounted for. Fig. 5 presents the workspace under both the tension and tipping conditions, where the lateral workspace is significantly reduced. This reduction arises because, as the EE moves laterally and approaches alignment with the front of a particular robot, the difference between the cable yaw angle and the mobile platform's yaw angle decreases. As a result, the cable-induced moment increases (as can be computed from Eq. (6)) and ultimately violates the tipping condition given by Eq. (4). Fig. 6 illustrates the workspace considering the tension and rotating conditions. In this case, the workspace decreases in both the lateral and front-rear directions, although the reduction mechanisms differ. The lateral reduction occurs because, as the EE moves sideways and approaches alignment with the front of a mobile platform, the yaw-direction moment generated by the cable tensions (Eq. (15)) increases, leading to rotational instability. The front-rear workspace decreases because, as the EE moves forward or backward and approaches the lateral side of a mobile platform, the moment generated by the cable tensions (Eq. (9)) increases, concentrating the supporting force on a single wheel, reducing the available frictional resistance on the other wheels, and consequently violating the rotating condition in Eq. (14).

The workspaces shown in Figs. 5 and 6 demonstrate that each condition affects the workspace in distinct ways. The presence of non-overlapping regions between the two workspaces indicates that both conditions must be considered simultaneously. Ultimately, when all conditions are taken into account, the

resulting workspace is as shown in Fig. 7.

4.2 Validation using simulation

To validate the algorithm, simulations were conducted using a model constructed in CoppeliaSim. The simulation followed the same conditions used in the algorithmic computations, with the end-effector moving to designated target points. The interaction between the end-effector and the mobile platforms due to the cables was implemented by applying tension vectors to the corners of the end-effector and the pulleys, based on the cable directions derived from the real-time kinematics. For intuitive comparison, the workspace was evaluated on a horizontal plane located at a height of 652 mm, which corresponds to approximately the mid-level of the workspace. When the end-effector moved outside the workspace defined by the tipping condition, tipping occurred, as shown in Fig. 8, and the resulting state is illustrated in Fig. 9. Additionally, when the end-effector moved outside the workspace defined by the rotating condition, rotating occurred as shown in Fig. 10, and the resulting state is depicted in Fig. 11.

4.3 Experimental validation

To further validate the algorithm and simulation, tension control experiments were conducted using a prototype MCDPR. The prototype setup is shown in Fig. 12.

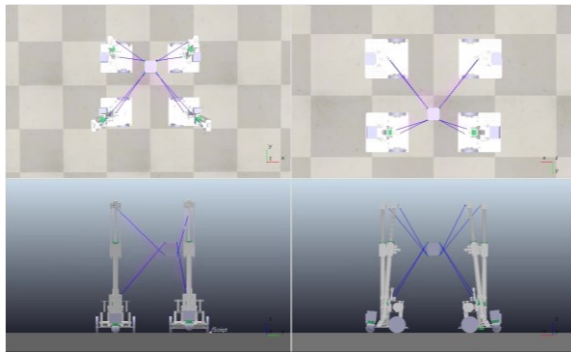


Fig. 8. Simulation of tipping situations using CoppeliaSim.

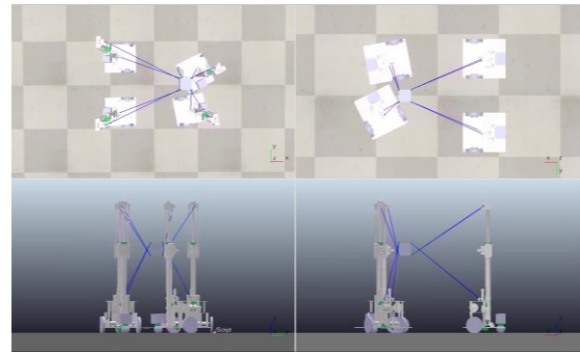


Fig. 10. Simulation of rotating situations using CoppeliaSim.

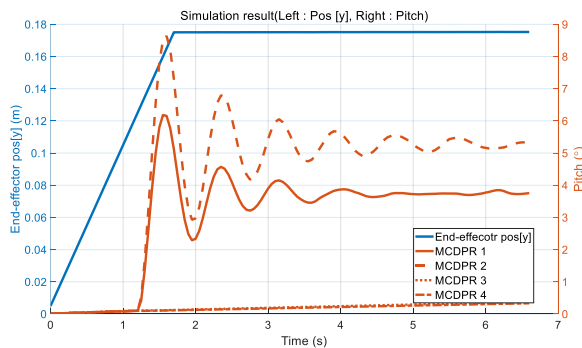


Fig. 9. Result of simulation (tipping).

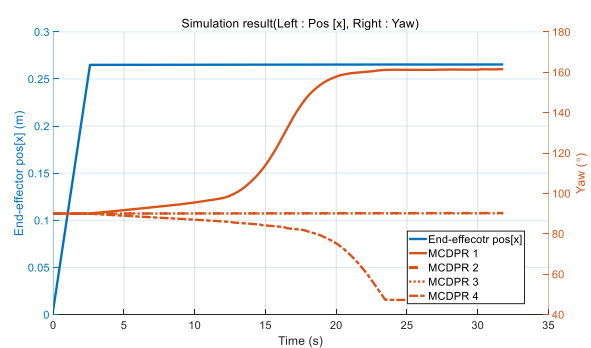


Fig. 11. Result of simulation (rotating).

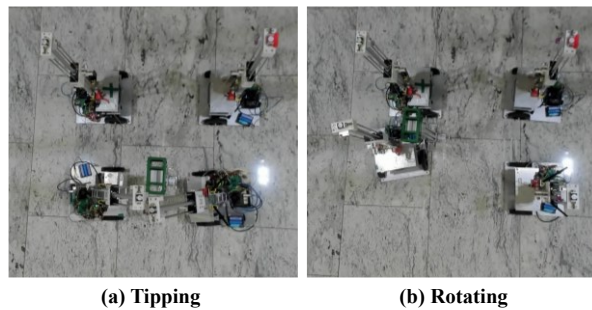


Fig. 12. Experiments with prototype MCDPR.

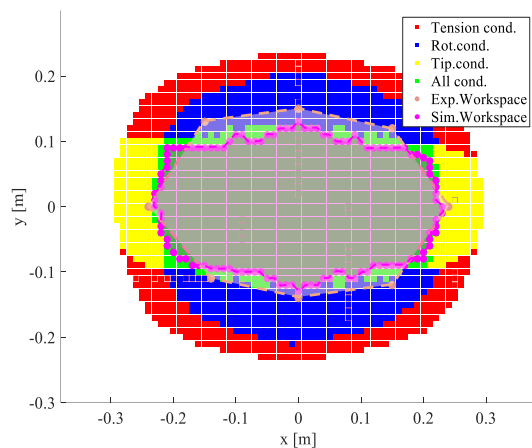


Fig. 13. Workspace in the earth environment.

The experiment was conducted under the same conditions as the simulation. After fixing the end-effector at specific target positions, tension control was performed using the tension sets calculated based on the workspace conditions satisfied at those positions. The theoretical workspace, the workspace obtained through simulation, and the workspace determined through experiments are shown in Fig. 13.

The comparison results of each workspace are summarized in Table 2.

The theory-based workspace and the simulation-based workspace exhibited a high similarity of over 91 %, confirming that the theoretical workspace can be effectively validated through simulation. The simulated workspace appeared approximately 8 % smaller than the theoretical workspace. This discrepancy may have arisen because, in the theoretical model, the wheel contact point was assumed to be located at the midpoint of the wheel thickness, whereas in the simulation's physics engine, the wheels made line contact or contacted at positions different from the theoretical assumption.

The theory-based workspace and the experiment-based workspace exhibited a high similarity of over 87 %, thereby confirming the validity of the theoretical workspace. In the experiments, the workspace appeared approximately 13 % larger than the theory-based workspace. This difference may stem from subtle variations in the coefficient of friction caused by dis-

Table 2. Comparison of workspaces.

Theory-based	Workspace	Overlap (%)	Relative area (%)
	Simulation-based	91.5	92.2
	Experiment-based	87.4	93.4

crepancies between the floor conditions during friction testing and those during the experiments, or from human-factor issues associated with setting the EE and mobile platforms position.

Additionally, minor limits of sensor (loadcell, IMU) resolution or time delays in the tension control loop may have contributed to the observed discrepancies.

5. Workspace of the MCDPR in lunar environment

To investigate the feasibility of operating the MCDPR in lunar environments, the theoretical workspace under lunar conditions (specifically, one-sixth of Earth's gravity and a friction coefficient of $\mu_{moon}=0.6$ [18]) was computed. It should be noted that the frictional characteristics of lunar regolith vary with particle size, compaction, and location, and thus the adopted value represents an average estimate reported in previous studies rather than a site-specific parameter. The results are shown in Table 3.

When comparing the workspaces on Earth and on the Moon, it was found that the workspace defined solely by the tension condition remained identical regardless of environmental differences. Although a significant reduction in the end-effector's weight can potentially decrease the workspace (since it may become difficult to compute a valid tension set that satisfies the minimum tension requirement) the simultaneous reduction in cable weight under lunar gravity offset this effect. As a result, the workspace defined solely by the tension condition remained unchanged. However, when both the tension condition and the tipping condition were considered, the lateral and upper regions of the workspace were reduced. This is attributed to the robot becoming more susceptible to tipping due to its decreased weight under lunar gravity. Similarly, when the tension condition and the rotating condition were considered, the upper workspace was reduced in the lunar environment. As with the tipping condition, the decrease in the robot's weight reduced the normal force on each wheel, making the robot more vulnerable to moment-induced load concentration, particularly in the upper region. As a result, when all conditions were considered, the lunar workspace exhibited a similar front-rear range compared to Earth, but the lateral and upper regions were notably reduced.

To validate the theoretical workspace, a simulation was conducted under lunar conditions using the same scenario as in the Earth-based simulation, implemented in CoppeliaSim. The simulation results are presented in Fig. 14.

The theory-based workspace and the simulation showed a 91.5 % overlap, with an area similarity of 92.2 %, confirming that the workspace computation algorithm remains effective under lunar conditions. Compared to the Earth-based workspace, the

Table 3. Workspace in the earth environment.

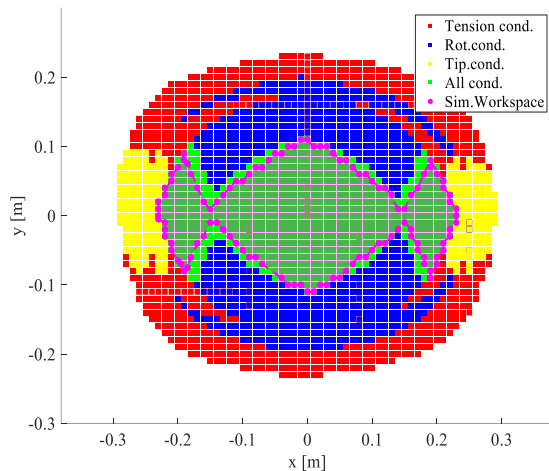
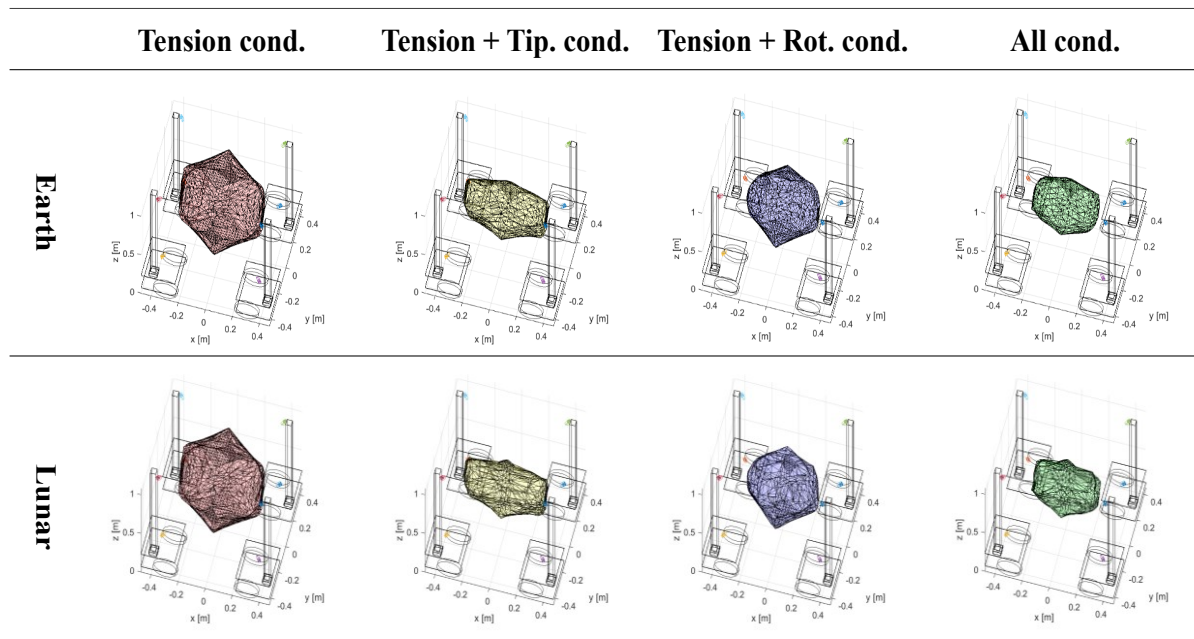


Fig. 14. Workspace in the lunar environment.

overall workspace decreased by 34 %. This reduction is attributed to the decrease in weight due to lower gravity, which weakens resistance against tipping moments. As a result, the workspace defined by the tipping condition tended to shrink in a concave pattern along the diagonal directions from the center toward each robot. Additionally, the reduced weight also decreased the available frictional force at the wheels, leading to crescent-shaped unusable regions in the front and rear areas of the workspace constrained by the rotating condition.

6. Conclusion

In this study, we proposed a workspace computation algorithm for the MCDPR (mobile cable-driven parallel robot) based

on static equilibrium conditions, and validated its effectiveness through simulations and experiments. The proposed algorithm incorporates tension, tipping, and rotating conditions, allowing for precise estimation of the robot's feasible operational space. It achieved over 87 % similarity with both simulation and experimental results. Applying the algorithm under lunar gravity and friction conditions confirmed that the resulting workspace was reduced by approximately 34 % compared to that on Earth. Notably, the workspace was constrained along the diagonal directions due to the tipping condition and along the front-rear directions due to the rotating condition. These findings provide practical foundational data for designing MCDPR platforms for autonomous additive construction in lunar environments.

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Conflict of interest

The authors declare no competing interests.

Nomenclature

- $\vec{a}_i$: The position vector of the i -th pulley
- $\vec{b}_i$: The position vector of the anchor point where the i -th cable is connected to the end-effector
- $d_{c,j}$: The distance on the contact plane between the contact

point of the caster wheel and the line segment connecting the contact points of the two drive wheels of the j -th robot

$d_{m,j}$: The distance between the center of mass of the j -th robot on the contact plane and the line segment connecting the contact points of the two drive wheels

$d_{m2w,j}$: The distance between the center of mass of the j -th robot on the contact plane and the contact point of the wheel where the supporting force is concentrated

$d_{p,i}$: The distance between the i -th pulley and the line segment connecting the two drive wheels

$d_{p2w,i}$: The distance on the contact plane between the position of the i -th pulley and the contact point of the wheel where the supporting force is concentrated

d_{w2w} : The distance between the two drive wheels

F : The external force matrix acting on the end-effector

$F_{c,j}$: The support force acting on the caster wheel of the j -th robot

$F_{n,j}$: The support force of the non-dominant drive wheel of the j -th robot

$F_{r,j}$: The virtual force applied at the center of mass that produces a moment equal to $M_{rot,j}$

g : The gravitational acceleration

$h_{p,i}$: The height of the i -th pulley

i : Number of cable

j : Number of robot

l_i : The length of the i -th cable

$\vec{l}_{i,xy}$: The xy -component of the unit direction vector of the i -th cable

$\vec{l}_{i,z}$: The z -component of the unit direction vector of the i -th cable

$M_{f,j}$: Moment resisting rotation due to friction

$M_{rc,i}$: The moment generated by the tension of the i -th cable for rotating condition

$M_{r,j}$: The total moment used to evaluate support concentration for the j -th robot

$M_{rm,j}$: The moment generated by the weight of the j -th robot for rotating condition

$M_{tc,i}$: The moment generated by the tension of the i -th cable for tipping condition

$M_{tm,j}$: The moment generated by the weight of the j -th robot for tipping condition

$M_{yc,i}$: The moment generated by the i -th cable about the axis perpendicular to the contact plane

m_j : The mass of the j -th robot

n : Degree of freedom

$\vec{p}_{EE}$: The target position vector of the end-effector

R_{EE} : The rotation matrix representing the target orientation of the end-effector

T : The matrix of cable tensions that satisfies the static equilibrium of the end-effector

T_h : The homogeneous tension matrix

T_i : The tension in the i -th cable

T_{min} : The minimum tension required to prevent the cable from sagging

T_{max} : The maximum allowable tension set to avoid damage to

the MCDPR

T_p : The particular tension matrix

T_{ref} : The reference tension matrix selected by considering physical constraints

W : The wrench matrix, which consists of vectors representing the forces and moments generated by the cables

W^+ : The Moore–Penrose pseudoinverse of W

$\gamma_{c,i}$: The yaw angle of the i -th cable

$\gamma_{m,j}$: The yaw angle of the j -th robot

μ : The coefficient of friction between the wheels and the ground

μ_{moon} : The coefficient of friction between the wheels and the ground of moon

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